Energy, Heat and Work

Answer key with worked solutions

A. Ideas

1. Definitions and systems.

> (a) **Energy**: the capacity to do work or transfer heat. **Heat**: energy transferred because of a temperature difference. **Work**: energy transferred by a force acting through a distance (e.g. a gas pushing a piston).
>
> (b) Boiling pan: **open** (steam and heat escape). Sealed balloon: **closed** (heat can pass, gas cannot). Insulated sealed flask: **isolated** (ideally, neither passes).

2. Bath and cup.

> The **cup** has the higher temperature (100 °C). The **bath** contains more thermal energy, because it has far more water: temperature reflects the **average** energy per particle, while the total thermal energy also depends on how many particles there are.

B. Kinetic and potential energy

3. Kinetic energy.

> (a) $E_\mathrm{k} = \frac{1}{2} \times 2.00\ \mathrm{kg} \times (3.00\ \mathrm{m/s})^2 = \mathbf{9.00\ J}$
>
> (b)
>
> $$E_\mathrm{k} = \tfrac{1}{2} \times 4.65 \times 10^{-26}\ \mathrm{kg} \times (515\ \mathrm{m/s})^2 = \mathbf{6.17 \times 10^{-21}\ J}$$

4. Potential energy.

> $E_\mathrm{p} = mgh = 5.00\ \mathrm{kg} \times 9.81\ \mathrm{m/s^2} \times 2.00\ \mathrm{m} = \mathbf{98.1\ J}$

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C. Energy units

5. Conversions.

> (a) $520\ \cancel{\text{kcal}} \times \dfrac{4.184\ \text{kJ}}{1\ \cancel{\text{kcal}}} = 2175.68\ \text{kJ} = \mathbf{2.18 \times 10^{3}\ kJ}$
>
> (b) $1200\ \cancel{\text{kJ}} \times \dfrac{1\ \text{kcal}}{4.184\ \cancel{\text{kJ}}} = \mathbf{287\ kcal}$
>
> (c) $2.5\ \cancel{\text{kWh}} \times \dfrac{3.6 \times 10^{6}\ \text{J}}{1\ \cancel{\text{kWh}}} = \mathbf{9.0 \times 10^{6}\ J}$

D. Work and the first law

6. Compression.

> $\Delta V = 4.0\ \text{L} - 10.0\ \text{L} = -6.0\ \text{L}$
>
> $$w = -P_{\text{ext}}\,\Delta V = -2.00\ \text{atm} \times (-6.0\ \text{L})$$
> $$= +12\ \cancel{\text{L atm}} \times \frac{101.325\ \text{J}}{1\ \cancel{\text{L atm}}}$$
> $$= \mathbf{+1.2 \times 10^{3}\ J}$$
>
> Positive: the surroundings do work **on** the gas, so the system gains energy.

7. ΔU.

> (a) $\Delta U = (-300.\ \text{J}) + (+150.\ \text{J}) = \mathbf{-150.\ J}$
>
> (b) $\Delta U = (+500.\ \text{J}) + (-200.\ \text{J}) = \mathbf{+300.\ J}$
>
> (c) Rigid container: $\Delta V = 0$, so $w = 0$ and $\Delta U = q = \mathbf{+75\ J}$

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E. Challenge

8. Piston.

(a) $\Delta V = 6.00\ \text{L} - 1.00\ \text{L} = 5.00\ \text{L}$

$$w = -1.00\ \text{atm} \times 5.00\ \text{L} = -5.00\ \cancel{\text{L atm}} \times \frac{101.325\ \text{J}}{1\ \cancel{\text{L atm}}} = \mathbf{-507\ J}$$

(b) $q = +2.50\ \text{kJ} = +2500\ \text{J}$

$$\Delta U = q + w = (+2500\ \text{J}) + (-507\ \text{J})$$

$$= +1993\ \text{J} = \mathbf{+1.99\ kJ}$$

(c) ΔU is positive, so the internal energy **increased**: the gas gained more energy as heat than it lost as work.

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